

Fluid Flow Through Fluidized Beds

Final Lab Report
Unit Operations Lab 2
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1. Executive Summary

This experiment investigated the fluidization behavior of sand and ion-exchange resin to determine the minimum fluidization velocity and to evaluate the relationship between pressure drop and superficial gas velocity. Compressed air was fed upward through packed beds of 6-inch and 10-inch heights for each material, and pressure drop was measured over a range of flow rates during both increasing and decreasing flow conditions. Superficial velocity was calculated from the volumetric flow rate and column cross-sectional area, and experimental results were compared to theoretical predictions based on the Ergun equation.

For silica sand, fluidization occurred at approximately 28 LPM in both 6-inch and 10-inch beds, corresponding to superficial velocities of approximately 0.059 m/s. The pressure drops at about 5.3 mm H₂O for the 6-inch bed and 11 mm H₂O for the 10-inch bed. The ion exchange resin required significantly higher flow rates to achieve fluidization, approximately 80-90 LPM for the 6-inch bed. With higher corresponding superficial velocities. Increased bed height resulted in greater pressure drop at fluidization for both materials, while minimum fluidization velocity remained relatively similar for a given particle type.

These results demonstrate that particle properties such as density and size strongly influence fluidization behavior. Silica sand fluidized at lower velocities due to lower particle density and drag requirements compared to the resin beads. The experiment confirms key trends in fluidization theory and highlights the importance of particle selection and bed height in industrial fluidized bed design.

2. Introduction (10 pts)

The objective of this experiment was to investigate gas-solid fluidization behavior in a laboratory-scale fluidized bed and to determine the minimum fluidization velocity for two different particulate materials: silica sand and ion-exchange resin beads. Fluidized beds are widely used in chemical and petrochemical industries for processes such as catalytic cracking, drying, combustion, adsorption, and polymerization because they provide excellent heat and mass transfer due to intense particle mixing. Understanding the conditions required to achieve fluidization is critical for proper reactor design, scale-up, and operation.

When a gas flows upwards through a packed bed of particles, the pressure drop initially increases linearly with superficial velocity while the bed remains fixed. As velocity increases further, the drag force exerted by the gas balances the particles' weight, and the bed transitions to a fluidized state. At this point, the pressure drop across the bed becomes approximately constant and equal to the solids' weight per unit cross-sectional area. The gas velocity at which this occurs is defined as the minimum fluidization velocity. The Ergun equation describes pressure drop in the fixed-bed regime and provides a theoretical basis for predicting fluidization behavior.

In this experiment, compressed air was introduced upward through a vertical column packed with either silica sand or ion exchange resin at bed heights of 6 inches and 10 inches. The airflow rate was varied incrementally in both increasing and decreasing directions, and the pressure drop across the bed was recorded. From these measurements, superficial velocity and minimum fluidization velocity were determined

and compared between materials and bed heights to evaluate the influence of particle properties and system geometry on fluidization performance.

3. Theory (10 pts)

When compressed air is fed into small particles sitting on top of each other, a fluidized bed can be created. To create a fluidized bed, the fluid, in this case compressed air, needs to reach a high enough velocity. Since the packed bed is composed of small particles, there is also a small spacing in between them. This is known as the void fraction, which can be calculated by dividing the volume of the void by the volume of the bed. Typically, the void fraction ranges from 0 to 1, where a zero void fraction would mean absolutely no free space in the bed (filled with sand or resin), and a void fraction of 1 would mean there's nothing blocking flow through the pipe. Therefore, as the void fraction increases, there is less resistance to flow, meaning it would reach fluidization more quickly. A major variable observed in fluidized beds is the pressure drop, which can change the behavior of the superficial velocity (velocity based on the bed's empty cross-section). At low gas velocities, the pressure drop through a bed can be described by the Ergun equation.

Where

- = pressure drop across the bed
- = height of the packed bed
- = the fluid density
- = is the superficial velocity
- = the bed voidage at minimum fluidization
- = the fluid viscosity
- = the sphericity
- = the particle diameter

Because not all particles have the same diameter, an approximation is used: particles are sieved, the mass fraction collected by each sieve is measured, and Equation 2 is used to find the average particle diameter.

Where X_i is the mass fraction of the particle of size D_{pi}

Throughout the experiment, the superficial velocity determines the fluidized bed phase and how the particles behave. At low velocity, the solid particles are moving at a minimum, and as velocity increases, they begin to bubble up; at this point, fluidization has been reached.

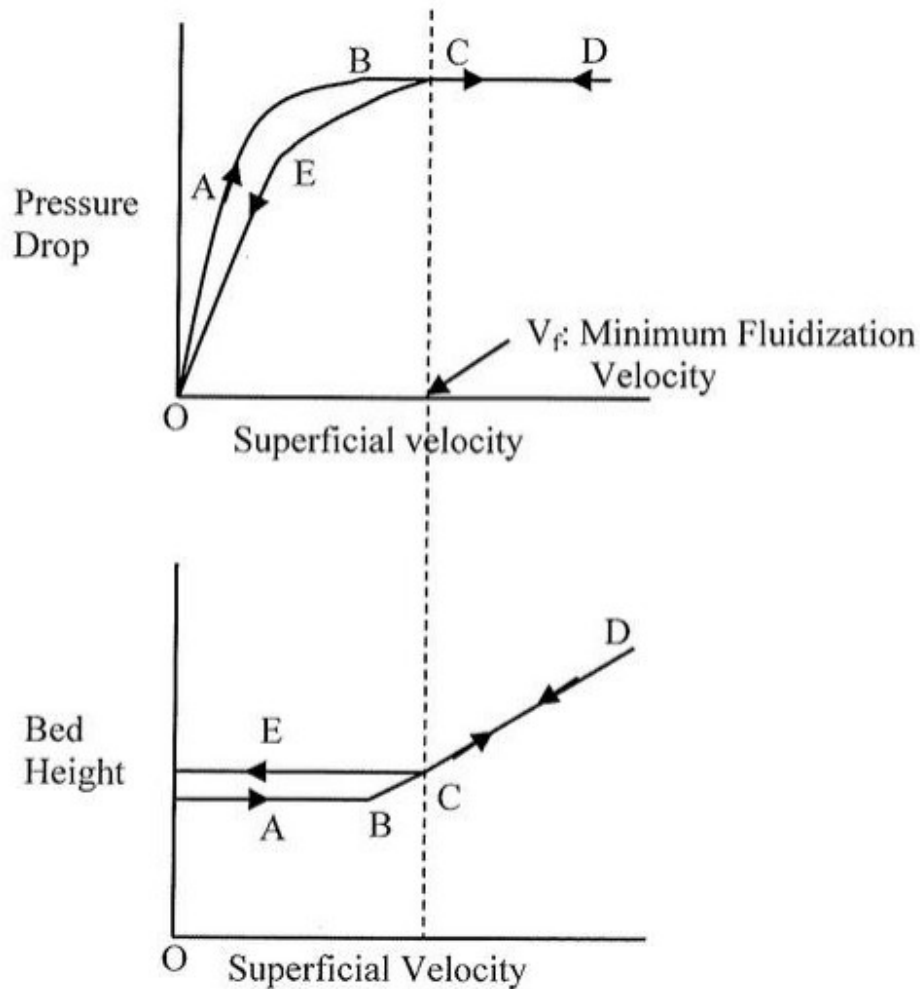


Figure 1: Pressure drop and bed porosity

Figure 1 shows that once fluidization is achieved, the pressure drop becomes constant.

4. Results (15 pts)

The objective of this experiment is to investigate the fluidization behavior of two different particles: silica sand and ion-exchange resin beads. For both materials, fluidization was tested at 6 and 10 inches. For both heights and materials, the pressure drop was measured at different air flow rates entering the fluidized bed. The minimum fluidization velocity was determined from the measured data. The pressure drop was then analyzed to determine when the bed reached fluidization and when a plateau was achieved.

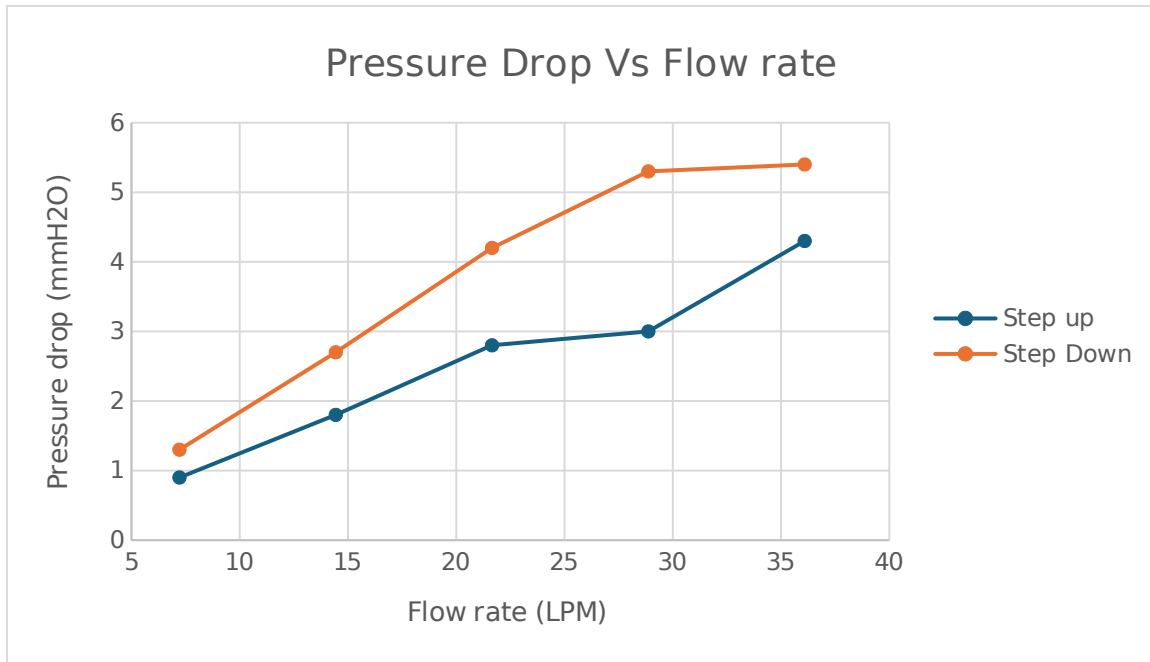


Figure 2: Pressure drop Vs Flow rate (sand 6in)

Figure 2 describes the relationship between pressure drop in mmH₂O vs. flow rate in LPM for a bed filled with 6 inches of sand. At a flow rate of about 28 LPM, the pressure drop appears to plateau at 5.3 mmH₂O, indicating fluidization has been achieved.

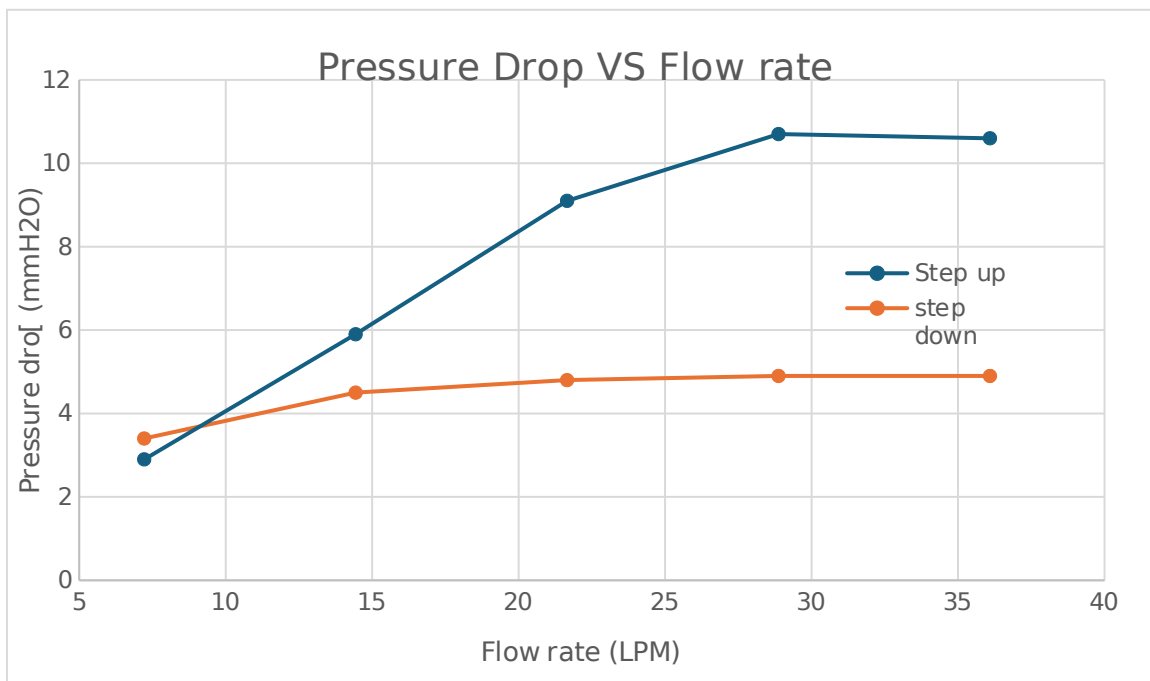


Figure 3: Pressure drop Vs Flow rate (sand 10in)

Figure 3 describes the relationship between pressure drop and flow rate of compressed air when the bed is filled with 10 inches of sand. A similar trend can be observed between figures 2 and 3, where in figure 3 the packed bed appears to fluidize at a flow rate of 28 LPM with a pressure drop of around 11 mmH₂O. The difference between 6 inches of sand and 10 inches of sand is that the latter requires a greater pressure drop to achieve fluidization, but about the same flow rate of compressed air.

Table 1: 6 inch Sand data (step up)

LPM	pressure drop/bed height (Pa/m)	Superficial Velocity (m/s)
7.2	57.9	0.0148
14.4	115.8	0.0296
21.7	180.2	0.044
28.9	193.1	0.059
36.1	276.7	0.0742

Table 2: 6 inch sand data (step down)

LPM	pressure drop/bed height (Pa/m)	Superficial Velocity (m/s)
36.1	315.3	0.07
28.9	315.3	0.059
21.7	308.9	0.044
14.4	289.6	0.029
7.2	218.8	0.014

Table 3: 10 inch sand data (step up)

LPM	Pressure drop (pa/m)	Superficial Velocity (m/s)
7.2	186.6	0.0148
14.4	379.7	0.0296
21.7	585.6	0.0445
28.9	688.5	0.0593
36.1	682.1	0.0742

Table 4: 10-inch sand data (step down)

LPM	Pressure drop (pa/m)	Superficial Velocity (m/s)
36.1	315.3	0.0742
28.9	315.3	0.0593
21.7	308.9	0.0445
14.4	289.6	0.0296
7.2	218.8	0.0148

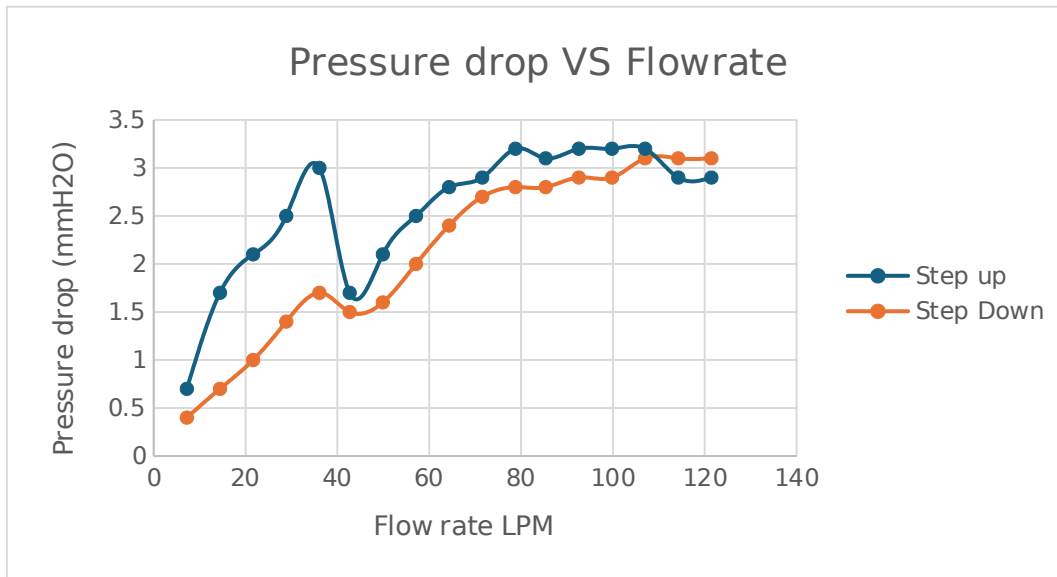


Figure 4: Pressure drop vs flow rate (Ion Resin) 6 inches

Figure 4 shows the relationship between the pressure drop and the flow rate for a fluidized bed filled with 6 inches of resin. From Figure 4, it can be inferred that in this system, at a flow rate of about 90 LPM, the resin is fluidized, as the pressure drop remains constant.

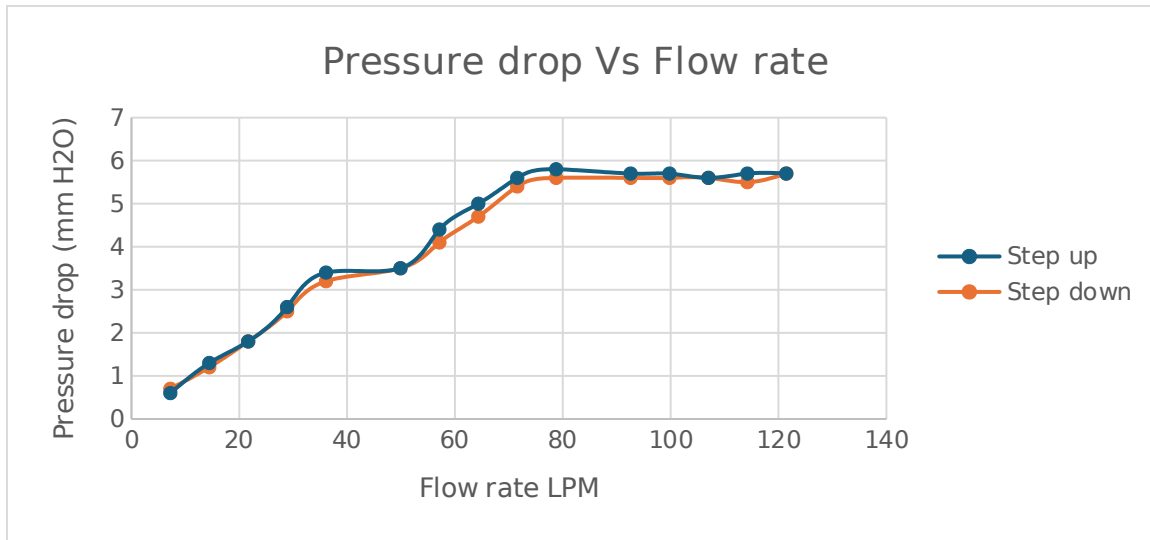


Figure 5: Pressure drop vs Flow rate (Ion Resin) 10 inches

Figure 6 shows the relationship between the pressure drop and the flow rate for a fluidized bed filled with 10 inches of Ion resin. Looking at figures 4 and 5, a similar trend is observed, with fluidization occurring at approximately the same flow rate. However, the main difference is in the pressure drop at which fluidization is achieved. For 6 inches of resin, the value is about 3 mm H₂O, and for 10 inches, it's about 5.5 mm H₂O. This is to be expected, as the longer the bed is, the higher the pressure drop.

Table 5: 6 inch Resin data (step up)

LPM	Pressure drop (pa/m)	Superficial Velocity (m/s)
7.2	45.1	0.014
14.4	109.4	0.03
21.6	135.1	0.044
28.9	160.9	0.059
36.1	193.1	0.074
42.7	109.4	0.087
49.9	135.1	0.103
57.1	160.8	0.117
64.3	180.2	0.132
71.6	186.6	0.147
78.8	205.9	0.162
85.4	199.5	0.176
92.6	205.9	0.190
99.8	205.9	0.205
107.1	205.9	0.220
114.2	186.6	0.235
121.5	186.6	0.250

Table : 6 inch Resin data (step down)

LPM	Pressure drop (pa/m)	Superficial Velocity (m/s)
121.5	199.48	0.249
114.2	199.48	0.234
107.1	199.48	0.220
99.8	186.61	0.205
92.6	186.61	0.190
85.4	180.17	0.176
78.8	180.17	0.162
71.6	173.74	0.147
64.3	154.43	0.132
57.1	128.69	0.117
49.9	102.95	0.102
42.7	96.52	0.088
36.1	109.39	0.074
28.9	90.1	0.059
21.7	64.35	0.044
14.4	45.04	0.029
7.2	25.74	0.015

Table 7: 10 inch Resin data (step up)

LPM	Pressure drop (pa/m)	Superficial Velocity (m/s)
7.2	38.6	0.0148
14.4	83.7	0.0296
21.7	115.8	0.0445
28.9	167.3	0.0593
36.1	218.8	0.0742
49.9	225.2	0.1026
57.1	283.1	0.1174
64.3	321.7	0.1323
71.6	360.3	0.1471
78.8	373.2	0.1620
92.6	366.8	0.1904
99.8	366.8	0.2053
107.2	360.3	0.2201
114.2	366.8	0.2349
121.5	366.8	0.2498

Table 8: 10-inch Resin data (step down)

LPM	Pressure drop (pa/m)	Superficial Velocity (m/s)
121.5	366.8	0.249
114.2	353.9	0.234
107	360.3	0.220
99.8	360.3	0.205
92.6	360.3	0.190
78.8	360.3	0.162
71.6	347.5	0.147
64.3	302.4	0.132
57.1	263.8	0.117
49.9	225.2	0.103
36.1	205.9	0.074
28.9	160.9	0.059
21.7	115.8	0.044
14.4	77.2	0.029
7.2	45	0.014

Ultimately, it can be concluded from the results that the packed bed with silica sand required less flowrate of compressed air to reach fluidization compared to that of a packed column with resin. For a bed packed with sand at 6 inches fluidization was achieved at a flowrate of roughly 28 LPM, while a bed packed with ion exchange resin fluidization was achieved at a flowrate of roughly 80 LPM.

Table 9: V_0 vs measured velocity (sand 6in)

6in Sand	
V_0 (m/s)	Measured Velocity (m/s)
3.16	0.0148
4.47	0.0296
5.57	0.0445
5.77	0.0593
6.91	0.0742

Table 10: V_0 vs measured velocity (sand 10 in)

Step up	
V_0 (m/s)	Measured Velocity (m/s)
0.009	0.0148
0.014	0.0296
0.017	0.0445
0.018	0.0593
0.018	0.0742

Table 11: V_0 vs measured velocity (sand 10 in)

step down	
V_0 (m/s)	Measured Velocity (m/s)
0.476	0.0742
0.481	0.0593
0.476	0.0445
0.476	0.0296
0.471	0.0148

Table 12: V_0 vs measured velocity (Resin 6 in)

step up	
V_0 (m/s)	Superficial Velocity (m/s)
0.131	0.014
0.204	0.029
0.227	0.044
0.248	0.059
0.271	0.074
0.204	0.087
0.227	0.102
0.248	0.117
0.262	0.132
0.267	0.147
0.280	0.162

Table 13: V_0 vs measured velocity (resin 6 in)

V_0 (m/s)	Superficial Velocity (m/s)
0.584	0.014
1.169	0.029
1.753	0.044
2.338	0.059
2.923	0.074
4.041	0.102
4.621	0.117
5.211	0.132
5.795	0.147
6.379	0.162

Discussion (15 pts)

After conducting the experiment, a few things can be noted about the behavior of the silica sand and ion exchange resin. The silica sand reached fluidization more quickly than the ion exchange resin; this may be due to differences in void fraction and particle density. In the lab, silica sand had a particle density of 0.0002 g/mL, and the ion-exchange resin had a density of 0.0006 g/mL. Sand with a lower particle diameter fluidizes more quickly. The correlation between particle density and fluidization behavior is that lower density corresponds to a lower weight per particle. Moreover, a lower density requires less drag force to suspend the fluid (air). Thus, it allows the sand to reach fluidization much more quickly than ion-exchange resin.

Secondly, when calculating the superficial velocity using the Ergun equation, sand shows much higher superficial velocities than ion-exchange resin. But when comparing 6 and 10 inches of sand, a marginal difference in superficial velocity is observed, with 10 inches showing velocities 3 times those of 6 inches. This can be explained by the greater pressure and velocity required to reach the fluidization point at the same particle diameter, hence requiring more flow along the bed. The same cannot be said for ion exchange resin, as a very small increase was observed from 6 to 10 inches. This can be explained by the resin being more difficult to fluidize, so a change in height did not increase the required fluidization velocity. Since ion exchange resins are highly

porous, additional airflow is needed to expand the bed rather than significantly increasing the fluidization velocity.

Additionally, when comparing the minimum fluidization velocity between sand and ion-exchange resin using the Ergun equation and equation 7.51, the same correlations observed in the superficial velocity were evident. Sand displayed a much higher minimum fluidization velocity than the ion-exchange resin, as explained in the prior section of the discussion.

Moreover, when analyzing the step-up and step-down measurements for 6 and 10-inch sand and ion-exchange resin, a hysteretic system is confirmed. Hysteresis is when the system shows different behavior as airflow increases versus decreases, as seen during step-up and step-down. During the step-down, the pressure drop didn't change significantly, unlike during the step-up. A significant change in pressure drop is observed at lower step-down values when fluidization in the bed is closer to 0. This trend is seen in both the sand and the ion-exchange resin. The time required for both to reach hysteresis varies significantly due to differences in void fraction and particle diameter. When calculating the superficial velocity for the step-down trials, the values are comparable to those for the step-up trials in the opposite direction. This is supported by the velocity effects being altered by the airflow rate, and by hysteresis not affecting the effective velocity of the fluidized bed.

6. References (10 pts)

Begin your References section here. Make sure that all references listed here are cited in the body of the report. For this lab course, at least 4 references are required, not including the Lab Manual. The American Chemical Society style guide is a commonly used reference format used in engineering reports.

Include a page break to start your Appendices.

7. Appendix I: Data Tabulation/Graphs (10 pts)

sand								
Sieve size	Mass of sieve(g)	sieve and sand (g)	mass in sieve (g)		xi	Dpi (m)	xi/Dpi	Dp, avg (m)
105	486.7	509.5	22.8		0.04	1.05E-04	369.98	1.97E-04
177	522.9	611.2	88.3		0.15	1.77E-04	850.01	
210	540.3	1014.3	474		0.81	2.10E-04	3845.87	
355	399	400.8	1.8		0.00	3.55E-04	8.64	
			586.9	sum	1.00		5074.50	
total sand	590.9	g						

ion exchange resin								
Sieve size	Mass of sieve(g)	sieve and resin(g)	mass in sieve (g)		xi	Dpi (m)	xi/Dpi	Dp, avg (m)
1000	617.7	619.2	1.5		0.007039	1.00E-03	7.04	5.83E-04
841	624.6	683.4	58.8		0.275927	8.41E-04	328.09	
595	594	714.1	120.1		0.563585	5.95E-04	947.20	
355	399.2	431.9	32.7		0.153449	3.55E-04	432.25	
			213.1	sum	1		1714.59	
total resin	215.6	g						

Sand						
	100mL grad cyl	with sand	sand and cyl	sand and water	xi	Void fraction
weight g)	46.4	74.6	121	139.4	0.868005739	0.131994261

Ionic Resin						
	25mL grad cyl	with resin	resin and cyl	sand and water	xi	Void fraction
weight g)	11.1	7.7	18.8	29.3	0.641638225	0.358361775

8. Appendix II: Error Analysis (10 pts)

- Quantitative

A point of error in the lab can be pointed to the measurement of the pressure drop. This is due to the precise leveling and calibration needed for accurate measurements. Moreover, mmH₂O must be converted to Pa to use the Ergun equation, and since mmH₂O is used for small flow and pressure systems, a misread pressure drop near minimum fluidization can significantly shift calculations and recorded fluidization of sand and ion exchange resin.

- Qualitative

6 inch sand ▾	6 inch ion exchange resin ▾
% difference	% difference
0.168875371	0.241434002
0.183104484	0.23314779
0.184746059	0.309606695
0.247576479	0.252764121
0.107677754	0.2478677
	0.062073128
	0.125924803
	0.106611281
	0.074261525
	0.035102158
	0.064228471

This error analysis displays the differences in the superficial velocity calculations between 6 inches of sand and 6 inches of ion exchange resin. This comparison was performed to show the percent difference in the values between the step-up and step-down analyses. It can be seen that differences of up to 25% in the sand and a 31% difference in the ion exchange resin.

9. Appendix III: Sample Calculations (10 pts)

Begin your Sample Calculations section here for one data set. State the data set that was used for the calculations, for instance; Trial #6: Parallel configuration, 8 LPM air flowrate. Include the equations used and numerical values for all calculations that you performed in the experiment.

To begin the calculations, the average particle size is calculated using the following formula:

Where x_i is the mass fraction and d_{pi} is the respective particle diameter. x_i is found by dividing the mass in sieve by the total mass collected. d_{pi} is given in microns so it was converted to meters.

Using data from sand:

x_i	d_{pi} (m)	x_i/d_{pi}	d_p, avg (m)
0.0388	0.0001	369.98	0.000197
48	05	27	064
0.1504	0.0001	850.00	
52	77	86	
0.8076	0.0002	3845.8	
33	1	73	
0.0030	0.0003	8.6393	
67	55	3	
sum	1	5074.5	04

The calculation from here will be using data from six in. of sand at the lowest flow rate. The flow rate is given in LPM, so it is converted to :

Also, the pressure drop is given in inch H_2O , so it is converted to Pa.

From here, the superficial velocity is found:

A: Area which is =

Now, the minimum fluidization can be calculated using the following formula:

Breakdown of known values:

Plugging all the values into the above equation provides:

From here, the theoretical pressure drop is calculated using the Ergun equation:

Plugging in all the known values gives:

10. Appendix IV: Individual Team Contributions

NAME: Adil Feratovic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended lab
Pre-Lab Editing	1	Edited procedure
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	3	Worked on this part
Project Safety Evaluation (prelab)	0	
Final Lab Editing	2	Edited parts worked on
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	

Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	1	Added used references
Data Tabulation/Graphs (final)	3	Worked on data tabulation
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	3	Did calculations
Power Point Presentation	1	
Total	19	

NAME: Aaron Casas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended lab sessions
Pre-Lab Editing		
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	.5	Completed safety portion of the safety prelab.
Final Lab Editing	2	Edited theory and results section.
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	2	Explained how and what equations were used to calculate the numerical values.
Results (final lab)	4	Included data and graph figures explaining the trends in data.
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	3	Calculated values, tabulated data, and formed graphs
Error Analysis (final lab)	0	

Sample Calculations (final Lab)	0	
Power Point Presentation	.5	Worked on safety and Challenges slide
Total	15	

NAME: Anna Cai

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended the CCP meeting
Pre-Lab Editing	1	
Experimental Objective (Prelab)	.5	
Apparatus (prelab)	1	Wrote descriptions
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	1	Fixed grammar
Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	1	Edited
Results (final lab)	3	Completed calculations
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	.5	
Data Tabulation/Graphs (final)	1	Formatted tables
Error Analysis (final lab)		
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Apparatus and materials
Total	16	

NAME: Hussein Rezk

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Operated the stirred tank
Pre-Lab Editing	1	Experimental
Experimental Objective (Prelab)	1	Experimental
Apparatus (prelab)		
Materials & Supplies (prelab)	1	Listed required material

Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	5	Discussed results and correlations
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	Conducted qualitative and quantitative analysis
Sample Calculations (final Lab)		
Power Point Presentation	2	Theory sections
Total	16	